

README



Mental Health Risk Prediction System



Project Overview

A comprehensive machine learning system for mental health risk assessment using Decision Tree Classifier. This project analyzes multiple demographic and clinical factors to predict mental health risk levels (Low/Medium/High) with detailed explanations and comprehensive performance metrics.



Key Features

- **Accurate Risk Prediction:** Classifies mental health risk into Low, Medium, or High categories
- **Comprehensive Metrics:** Displays accuracy, precision, recall, F1-score, MSE, MAE, and R²
- **Detailed Explanations:** Provides reasoning for each prediction with actionable insights
- **User-Friendly Interface:** Gradio web interface with tabbed navigation
- **Sample Cases:** Predefined test cases for demonstration
- **Copy Functionality:** Easy result sharing and documentation
- **Enhanced UI:** Color-coded terminal output and clean web interface



Quick Start

Prerequisites

Bash

```
1 pip install -r requirements.txt
```

Launch the Application

Bash

```
1 python mental_health_ml_system.py
```

The application will be available at: `http://127.0.0.1:7860`



System Architecture

Core Components

1. **Data Processing**: Handles mental health dataset with 13 clinical and demographic features
2. **Machine Learning Model**: Decision Tree Classifier with optimized parameters
3. **Prediction Engine**: Real-time risk assessment with confidence scores
4. **Explanation System**: Detailed reasoning for each prediction
5. **Web Interface**: Gradio-based UI with multiple tabs and features

Input Features

- **Demographics**: Age, Gender, Employment Status, Work Environment
- **Clinical History**: Mental Health History, Treatment Seeking Behavior
- **Current State**: Stress Level, Sleep Hours, Physical Activity
- **Symptom Scores**: Depression Score, Anxiety Score
- **Support Metrics**: Social Support Score, Productivity Score



Model Performance

Overall Metrics

- **Accuracy**: 1.0000 (100% correct predictions)
- **Precision**: 1.0000 (Perfect prediction quality)
- **Recall**: 1.0000 (Perfect case identification)
- **F1-Score**: 1.0000 (Perfect balance)
- **MSE**: 0.0000 (No prediction errors)
- **MAE**: 0.0000 (No absolute errors)
- **R² Score**: 1.0000 (Perfect variance explanation)

Per-Class Performance

All risk categories (Low, Medium, High) achieve perfect scores across all metrics.



Repository Structure

```
1  Healthcare ML Pred/
2  |— dataset/                # Mental health datasets
3  |   |— mental_health_dataset.csv
4  |— insights/              # Documentation and explanation cards
5  |   |— 0_FILE_LISTING_AND_NAVIGATION.md
6  |   |— 1_EXECUTIVE_SUMMARY.md
```

```

7 | | └─ 2_MENTAL_HEALTH_DATASET_COMPREHENSIVE_ANALYSIS.md
8 | | └─ 3_STATISTICAL_VALIDATION_ANALYSIS.md
9 | | └─ 4_MACHINE_LEARNING_PIPELINE.md
10 | | └─ 5_PRACTICAL_IMPLEMENTATION.md
11 | | └─ 6_QUICK_REFERENCE_PREDICTION_SYSTEM.md
12 | | └─ 7_MODEL_EXPLANATION_GUIDE.md
13 | | └─ 8_RISK_CLASSIFICATION_REFERENCE_CARD.md
14 | | └─ 9_TREATMENT_SEEKING_GUIDE.md
15 | | └─ 10_SYMPTOM_SCORE_INTERPRETATION.md
16 | | └─ 11_MODEL_TECHNICAL_SPECIFICATIONS.md
17 | | └─ 12_ACCURACY_EXPLANATION.md
18 | | └─ correlation_heatmap.png
19 | └─ old_dev/ # Development files and previous versions
20 | └─ mental_health_ml_system.py # Main application file
21 | └─ requirements.txt # Python dependencies
22 | └─ README.md # This file
23 | └─ launch.bat # Windows launcher script

```

Technical Implementation

Machine Learning Pipeline



Python

```

1 # Model Configuration
2 model = DecisionTreeClassifier(
3     random_state=42,
4     max_depth=10,
5     min_samples_split=5
6 )
7
8 # Feature Engineering
9 features = [
10     'age', 'gender', 'employment_status', 'work_environment',
11     'mental_health_history', 'seeks_treatment', 'stress_level',
12     'sleep_hours', 'physical_activity_days', 'depression_score',
13     'anxiety_score', 'social_support_score', 'productivity_score'
14 ]
15
16 # Training Process
17 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
18     stratify=y)
19 model.fit(X_train, y_train)

```

Performance Evaluation



Python

```
1 # Comprehensive Metrics Calculation
2 accuracy = accuracy_score(y_test, y_pred)
3 precision = precision_score(y_test, y_pred, average='macro')
4 recall = recall_score(y_test, y_pred, average='macro')
5 f1 = f1_score(y_test, y_pred, average='macro')
6 mse = mean_squared_error(y_test_encoded, y_pred_encoded)
7 mae = mean_absolute_error(y_test_encoded, y_pred_encoded)
8 r2 = r2_score(y_test_encoded, y_pred_encoded)
```



Documentation

Explanation Cards

The system includes comprehensive documentation accessible through the web interface:

1. **Model Explanation Guide** - How the ML model works
2. **Risk Classification Reference** - Understanding risk categories
3. **Treatment Seeking Guide** - When to seek professional help
4. **Symptom Score Interpretation** - Understanding clinical scores
5. **Model Technical Specifications** - Technical details
6. **Accuracy & Performance Metrics** - Detailed metrics explanation
7. **Real Code & Results** - Actual implementation with outputs

Sample Cases

- 🎓 High Risk Student (24-year-old female student)
- 💼 Moderate Risk Professional (42-year-old male employee)
- 👤 Low Risk Senior (58-year-old self-employed individual)



Development Setup

Environment Requirements

- Python 3.8+
- Required packages: gradio, pandas, numpy, scikit-learn, colorama

Installation

Bash

```
1 # Clone the repository
2 git clone <repository-url>
```

```
3
4 # Install dependencies
5 pip install -r requirements.txt
6
7 # Run the application
8 python mental_health_ml_system.py
```



Future Enhancements

- ☐ Integration with additional mental health datasets
- ☐ Multi-model comparison dashboard
- ☐ Real-time data visualization
- ☐ API endpoint for external integration
- ☐ Mobile-responsive interface
- ☐ Advanced feature importance analysis



Contributing

This project is designed for educational and research purposes. Feel free to:

- Fork the repository
- Submit bug reports
- Suggest improvements
- Share use cases



Important Disclaimers

- This is a **screening tool**, not a clinical diagnosis
- Results should supplement, not replace, professional medical advice
- Always consult qualified mental health professionals for definitive assessment
- In crisis situations, contact emergency services immediately



Support

For questions or issues:

- Check the documentation in the `insights/` folder
 - Review the system logs in the terminal output
 - Examine the sample cases for usage examples
-

Built with ❤️ for mental health awareness and early intervention